Q1 JEE Main 2020 - 7 Jan (Morning)

If $g(x) = x^2 + x - 1$ and $(gof)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to.

(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) $-\frac{1}{3}$
(D) $\frac{1}{3}$

Q2 JEE Main 2020 - 8 Jan (Morning)

The inverse function of $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$, $x \in (-1,\ 1)$, is

(A) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$
(B) $\frac{1}{2}\log_8\left(\frac{1-x}{1+x}\right)$
(C) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1-x}{1+x}\right)$
(D) $\frac{1}{2}\log_8\left(\frac{1+x}{1-x}\right)$

Q3 JEE Main 2020 - 8 Jan (Evening)

Let $f : (1,\ 3) \to R$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$. Then
the range of f is :

(A) $\left(0,\ \frac{1}{2}\right) \cup \left(\frac{3}{5},\ \frac{7}{5}\right]$
(B) $\left(\frac{2}{5},\ \frac{1}{2}\right) \cup \left(\frac{3}{5},\ \frac{4}{5}\right]$
(C) $\left(\frac{2}{5},\ 1\right) \cup \left(1,\ \frac{4}{5}\right]$
(D) $\left(0,\ \frac{1}{3}\right) \cup \left(\frac{2}{5},\ \frac{4}{5}\right]$

Q4 JEE Main 2020 - 9 Jan (Morning)

The number of distinct solutions of the equation, $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$ in the interval $[0,\ 2\pi]$, is

Q5 JEE Main 2020 - 9 Jan (Evening)

Let f and g be differentiable functions on R such that fog is the identity function. If for some $a,\ b \in R,\ g'(a) = 5$ and $g(a) = b$, then $f'(b)$ is equal to :

(A) $\frac{1}{5}$
(B) $\frac{2}{5}$
(C) 1
(D) 5

Q6 JEE Main 2020 - 2 Sep (Morning)

The domain of the function $f(x) = \sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty, -\mathrm{a}] \cup [\mathrm{a}, \infty)$. Then a is equal to :

(A) $\frac{1+\sqrt{17}}{2}$
(B) $\frac{\sqrt{17}}{2} + 1$
(C) $\frac{\sqrt{17}-1}{2}$
(D) $\frac{\sqrt{17}}{2}$

Q7 JEE Main 2020 - 2 Sep (Evening)

Let $f: R \to R$ be a function which satisfies
$f(x+y) = f(x) + f(y) \forall x, y \in R$. If $f(1) = 2$ and $g(n) = \sum_{k=1}^{(n-1)} f(k), n \in N$ then the value of n, for which $g(n) = 20$, is :

(A) 20
(B) 9
(C) 5
(D) 4

Q8 JEE Main 2020 - 4 Sep (Morning)

Let [t] denote the greatest integer $\leq t$. Then the equation in x, $[x]^2 + 2[x+2] - 7 = 0$ has :

(A) exactly two solutions.
(B) infinitely many solutions.
(C) exactly four integral solutions.
(D) no integral solution.

Q9 JEE Main 2020 - 6 Sep (Morning)

If $f(x+y)=f(x)f(y)$ and $\sum_{x=1}^{\infty} f(x)=2, x, y \in N$ where N is the set of all natural numbers, then the value of $\frac{f(4)}{f(2)}$ is:

(A) $\frac{1}{9}$
(B) $\frac{4}{9}$
(C) $\frac{1}{3}$
(D) $\frac{2}{3}$

Q10 JEE Main 2020 - 6 Sep (Evening)

For a suitably chosen real constant a, let a function, $f: R-\{-a\} \rightarrow R$ be defined by $f(x)=\frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and $f(x) \neq -a, (fof)(x)=x$. Then $f\left(-\frac{1}{2}\right)$ is equal to

(A) -3
(B) $\frac{1}{3}$
(C) $-\frac{1}{3}$
(D) 3

Q11 JEE Main 2020 - 6 Sep (Evening)

Suppose that a function $f: R \rightarrow R$ satisfies $f(x+y)=f(x)f(y)$ for all $x, y \in R$ and $f(1)=3$. If $\sum_{i=1}^{n} f(i)=363$, then n is equal to

Answer Key

Q1 (B)	**Q2** (A)	**Q3** (B)	**Q4** (8)
Q5 (A)	**Q6** (A)	**Q7** (C)	**Q8** (B)
Q9 (B)	**Q10** (D)	**Q11** (5)	

Q1

$$g\left(f\left(x\right)\right)=f^{2}\left(x\right)+f\left(x\right)-1$$

$$g\left(f\left(\frac{5}{4}\right)\right)=4\left(\frac{5}{4}\right)^{2}-10\cdot\frac{5}{4}+5=-\frac{5}{4}$$

$$g\left(f\left(\frac{5}{4}\right)\right)=f^{2}\left(\frac{5}{4}\right)+f\left(\frac{5}{4}\right)-1$$

$$-\frac{5}{4}=f^{2}\left(\frac{5}{4}\right)+f\left(\frac{5}{4}\right)-1$$

$$f^{2}\left(\frac{5}{4}\right)+f\left(\frac{5}{4}\right)+\frac{1}{4}=0$$

$$\left(f\left(\frac{5}{4}\right)+\frac{1}{2}\right)^{2}=0$$

$$f\left(\frac{5}{4}\right)=-\frac{1}{2}$$

Q2

$$y=\frac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}$$

$$\frac{1+y}{1-y}=\frac{8^{2x}}{8^{-2x}}$$

$$8^{4x}=\frac{1+y}{1-y}$$

$$4x=\log_{8}\left(\frac{1+y}{1-y}\right)$$

$$x=\frac{1}{4}\log_{8}\left(\frac{1+y}{1-y}\right)$$

$$f^{-1}\left(x\right)=\frac{1}{4}\log_{8}\left(\frac{1+x}{1-x}\right)$$

Q3

$$f(x)=\begin{cases}\frac{x}{x^2+1}; & x\in(1,2)\\ \frac{2x}{x^2+1}; & x\in[2,3)\end{cases}$$

$\therefore \quad f(x)$ is a decreasing function

$$\therefore \quad y\in\left(\frac{2}{5},\frac{1}{2}\right)\cup\left(\frac{6}{10},\frac{4}{5}\right]$$

$$\Rightarrow \quad y\in\left(\frac{2}{5},\frac{1}{2}\right)\cup\left(\frac{3}{5},\frac{4}{5}\right]$$

Q4

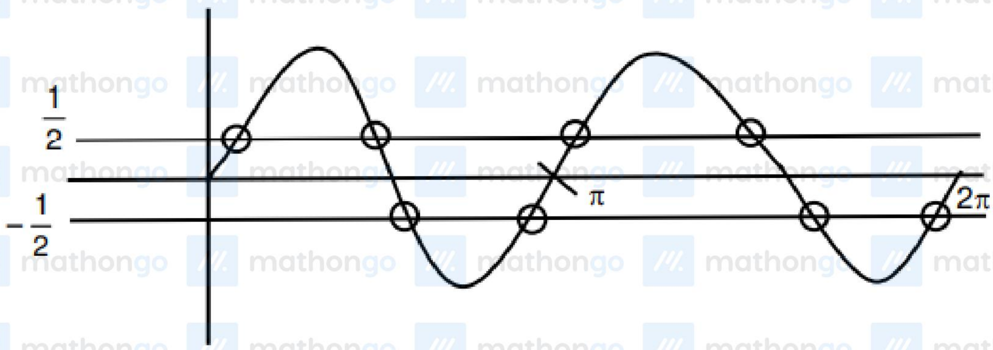
$\log_{1/2}|\sin x|=2-\log_{1/2}|\cos x|$

$\log_{1/2}|\sin x\cos x|=2$

$|\sin x\cos x|=\frac{1}{4}$

$\sin 2x=\pm\frac{1}{2}$

Number of solution $=8$.



Q5

$f(g(x))=x$

$\Rightarrow \quad f'(g(x))\cdot g'(x)=1$

Put $x=a$

$\Rightarrow \quad f'(g(a))\,g'(a)=1$

$\Rightarrow \quad f'(b)\times 5=1$

$\Rightarrow \quad f'(b)=\frac{1}{5}$

Q6

$\because f(x)=\sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$
$\therefore -1\leq\frac{|x|+5}{x^2+1}\leq 1$
$\therefore x^2-|x|-4\geq 0$
$\therefore |x|-\frac{1-\sqrt{17}}{2}\mid\left(|x|-\frac{1+\sqrt{17}}{2}\right)\geq 0$
$\therefore x\in\left(-\infty,-\frac{1+\sqrt{17}}{2}\right]\cup\left[\frac{1+\sqrt{17}}{2},\infty\right)$

Q7

$f(x+y)=f(x)+f(y),\quad \forall x,y\in R, f(1)=2$
$\Rightarrow f(x)=2x$
Now, $g(n)=\sum_{k=1}^{n-1}f(k)$
$=f(1)+f(2)+f(3)+\ldots\ldots f(n-1)$
$=2+4+6+\ldots\ldots+2(n-1)$
$=2[1+2+3+\ldots\ldots+(n-1)]$
$=2\times\frac{(n-1)(n)}{2}=n^2-n$
So, $n^2-n=20$ (given)
$\Rightarrow n^2-n-20=0$
$(n-5)(n+4)=0$
$\Rightarrow \quad n=5$

Q8

$[x]^2+2[x]+4-7=0$
$\Rightarrow [x]^2+2[x]-3=0$
$\Rightarrow [x]^2+3[x]-[x]-3=0$
$\Rightarrow ([x]+3)([x]-1)=0$
$\Rightarrow [x]=1$ or -3
$\Rightarrow \quad x\in[-3,-2)\cup[1,2)$
infinitely many solutions.

Q9

Let $f(1)=a$
then $f(1+1)=a^2$
$f(2+1)=a^3$

and so on.

$\sum_{x=1}^{\infty} f(x) = 2 \Rightarrow a + a^2 + a^3 + \ldots\ldots\ldots \infty = 2$

$\Rightarrow \frac{a}{1-a} = 2$

$\Rightarrow \quad a = \frac{2}{3}$

Now, $\frac{f(4)}{f(2)} = \frac{a^4}{a^2} = a^2 = \frac{4}{9}$

Q10

$f(f(x)) = \frac{a-\left(\frac{a-x}{a+x}\right)}{a+\left(\frac{a-x}{a+x}\right)} = x$

$\Rightarrow \frac{a^2+ax-a+x}{a^2+ax+a-x} = x$

$\Rightarrow a^2 + (a+1)x - a = a^2x + (a-1)x^2 + ax$

$\Rightarrow (a-1)x^2 + \left(a^2-1\right)x + \left(a-a^2\right) = 0$

$\forall x \in R - \{-a\}$

Hence $a = 1$

$\therefore \quad f(x) = \frac{1-x}{1+x} \quad \Rightarrow \quad f\left(-\frac{1}{2}\right) = 3$

Q11

$\because f(x+y) = f(x).f(y) \quad \forall x \in R \quad f(1) = 3$

$\Rightarrow \quad f(x) = 3^x \Rightarrow f(i) = 3^i$

$\Rightarrow \sum_{i=1}^{n} f(i) = 363$

$\Rightarrow 3 + 3^2 + 3^3 + \ldots + 3^n = 363$

$3^n - 1 = \frac{363 \times 2}{3} = 242$

$3^n = 243 = 3^5$

$n = 5$